Take this puzzle that was presented: A dealer in antique coins gets an offer to buy a beautiful bronze coin. The coin has an emperor's head on one side and the date "544 B.C." stamped on the other. The dealer examines the coin, but instead of buying it, he calls the police. Why?

Solving this "insight problem" requires creativity, a skill at which humans excel (the coin is a fake – "B.C." and Arabic numerals didn't exist at the time) and computers do not. But a new explanation of how humans solve problems creatively – including the mathematical formulations for facilitating the incorporation of the theory in artificial intelligence programs – provides a roadmap to building systems that perform like humans at the task.

Ron Sun, Rensselaer Polytechnic Institute professor of cognitive science, said his new "Explicit-Implicit Interaction" theory could be used for future artificial intelligence.

"As a psychological theory, this theory pushes forward the field of research on creative problem solving and offers an explanation of the human mind and how we solve problems creatively," Sun said. "But this model can also be used as the basis for creating future artificial intelligence programs that are good at solving problems creatively."

The paper, titled *Incubation, Insight, and Creative Problem Solving: A Unified Theory and a Connectionist Model,* by Sun and Sèbastien Hèlie of University of California, Santa Barbara, appeared in the July edition of Psychological Review. Discussion of the theory is accompanied by mathematical specifications for the "CLARION" cognitive architecture – a computer program developed by Sun's research group to act like a cognitive system – as well as successful computer simulations of the theory.

In the paper, Sun and Hèlie compared the performance of the CLARION model using Explicit-Implicit Interaction theory with results from previous human trials – including tests involving the coin question – and found results to be nearly identical in several aspects of problem solving.

In the tests involving the coin question, human subjects were given a chance to respond after being interrupted either to discuss their thought process or to work on an unrelated task. In that experiment, 35.6 per cent of participants answered correctly after discussing their thinking, while 45.8 per cent of participants answered correctly after working on another task.